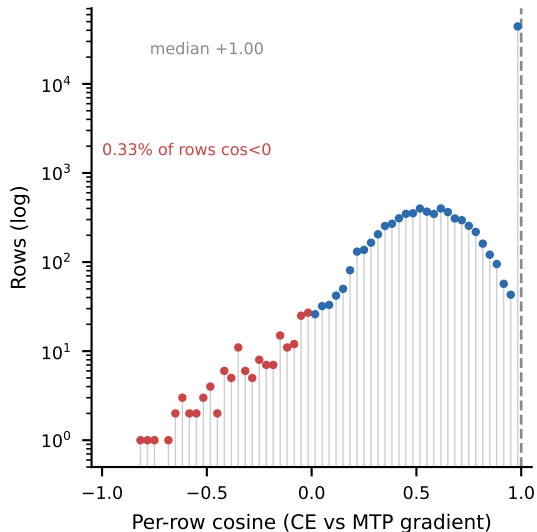
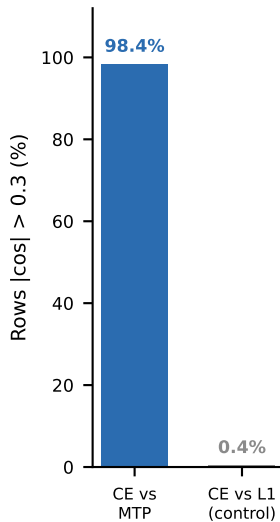


Per-row gradients align in direction



Instrument calibrates



- aligned ( $\cos \geq 0$ )
- opposed ( $\cos < 0$ )

GPT-2 124M, step 1000  
50,257 vocab rows  
(47 padding masked)